

JOHNSON COUNTY EXECUTIVE, KS, USA

WMO: 724468

Lat: 38.850N

Lon: 94.739W

Elev: 326

StdP: 97.47

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 03967

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.5	-12.7	-20.6	0.6	-14.0	-18.2	0.8	-11.4	11.2	2.3	10.3	3.4	3.8	330	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	35.2	24.3	33.3	24.2	32.1	23.9	26.0	31.9	25.4	31.2	24.7	30.2	4.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	20.0	29.5	23.7	19.3	28.9	22.9	18.3	28.0	82.5	31.9	79.6	31.1	76.6	30.2	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	8.9	8.1	-19.4	37.5	2.5	2.2	-21.2	39.0	-22.7	40.3	-24.1	41.5	-25.9	43.1		
			-19.8	27.3	2.4	0.6	-21.5	27.8	-22.9	28.2	-24.3	28.5	-26.0	29.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	-0.8	1.5	7.4	13.0	18.3	23.5	26.0	25.1	20.7	14.0	7.1	1.0
	DBStd	10.60	6.52	6.35	6.17	5.15	4.27	3.17	2.99	3.12	3.98	5.00	5.60	5.97
	HDD10.0	1155	337	246	122	28	2	0	0	0	0	19	119	283
	HDD18.3	2679	592	472	341	169	54	3	0	1	21	151	338	537
	CDD10.0	2300	3	7	42	119	260	404	494	469	322	143	32	4
	CDD18.3	781	0	0	2	10	54	158	236	211	93	16	1	0
	CDH23.3	7304	0	1	12	79	407	1406	2489	2041	753	116	2	0
	CDH26.7	2868	0	0	1	12	93	498	1105	884	251	24	0	0
Wind	WSAvg	3.9	4.2	4.3	4.6	4.7	4.0	3.8	3.2	3.1	3.4	3.8	4.1	4.1
Precipitation	PrecAvg	1012	32	32	65	95	131	146	101	98	121	85	66	39
	PrecMax	1630	84	100	264	257	314	340	396	231	372	215	216	124
	PrecMin	580	1	0	15	12	22	31	7	11	13	6	0	0
	PrecStd	227	22	24	46	51	67	78	74	58	93	56	47	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.5	20.8	25.0	28.2	31.1	34.7	37.5	37.6	34.1	29.1	22.9	18.3
		MCWB	11.8	12.7	16.6	18.7	21.8	24.4	24.1	23.5	22.6	21.0	16.1	14.8
	2%	DB	14.5	17.2	22.3	25.8	29.0	32.5	35.6	35.0	31.4	26.5	20.8	15.4
		MCWB	9.6	11.6	15.3	17.7	20.9	24.0	24.3	24.4	22.7	19.0	15.0	11.9
	5%	DB	11.7	13.9	19.8	23.5	27.5	31.2	33.7	33.0	29.4	23.9	17.9	12.3
		MCWB	7.3	9.1	14.0	16.6	20.3	23.6	24.6	23.8	22.3	17.9	12.9	8.7
	10%	DB	8.3	11.2	17.4	21.7	25.8	29.5	32.2	31.3	27.6	22.2	15.9	9.6
		MCWB	4.8	7.2	12.3	15.8	19.5	22.7	24.5	23.5	21.1	17.2	11.6	6.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.0	15.3	18.5	20.6	23.8	26.2	27.3	26.8	25.1	22.5	17.9	15.7
		MCDB	16.1	18.5	23.3	26.6	29.1	31.8	32.8	32.8	30.2	26.8	20.9	17.6
	2%	WB	10.3	12.4	16.8	19.2	22.5	25.3	26.3	25.7	24.0	21.0	15.7	12.5
		MCDB	13.6	16.0	20.6	23.5	27.2	30.9	32.2	31.8	29.0	24.6	19.0	14.6
	5%	WB	7.4	9.8	15.0	17.9	21.4	24.4	25.6	25.0	23.1	19.5	14.1	9.2
		MCDB	10.8	13.4	18.8	22.0	25.8	29.3	31.5	31.0	28.2	22.6	16.9	11.5
	10%	WB	5.0	7.1	12.6	16.4	20.3	23.5	25.0	24.2	22.1	17.6	12.1	6.7
		MCDB	8.6	10.9	16.6	20.9	24.5	28.3	30.6	29.6	26.5	20.5	15.3	9.4
Mean Daily Temperature Range	5% DB	MDBR	9.5	9.8	10.6	10.5	10.0	9.7	9.9	10.2	10.5	10.4	9.8	9.1
		MCDBR	13.7	14.5	13.9	12.9	11.6	11.0	11.8	12.4	11.8	12.3	12.6	12.5
		MCWBR	9.4	9.7	8.3	7.6	6.0	4.9	4.2	4.7	5.3	6.6	8.1	9.1
	5% WB	MCDBR	12.5	13.1	11.3	11.1	10.4	10.1	10.3	10.8	10.3	10.2	10.9	11.4
		MCWBR	9.4	9.8	7.9	7.8	6.1	5.4	4.7	4.9	5.2	6.6	8.3	9.5
Clear-Sky Solar Irradiance	taub	0.290	0.303	0.333	0.381	0.423	0.425	0.434	0.434	0.385	0.341	0.302	0.287	
	taud	2.478	2.452	2.424	2.304	2.245	2.270	2.271	2.273	2.364	2.474	2.527	2.513	
	Ebn at Noon	892	927	927	895	860	855	844	835	859	869	872	872	
	Edh at Noon	79	92	106	127	137	134	133	130	111	90	75	71	
All-Sky Solar Radiation	RadAvg	2.23	3.00	3.90	4.96	5.70	6.47	6.44	5.72	4.79	3.43	2.43	1.93	
	RadStd	0.23	0.34	0.33	0.32	0.51	0.48	0.40	0.31	0.32	0.39	0.24	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page